



BUSINESS BLUEPRINT

AOP AUTOMATION FOR VMD VCM PLANT

Honeywell

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Business Blueprint

AOP Automation for VMD VCM Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

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Revision History

Rev.	Date	Originated By	Description of status and changes
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1. Introduction

The purpose of this document is to capture detailed specifications for the Digitalized Annual Operating Plan (AOP) for VCM plant for VMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various VCM plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS or SAP and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for VCM technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
VMD VCM DCT	-	Digital AOP VMD VCM DCT	 VMD VCM DCT Rev 00.xlsx

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

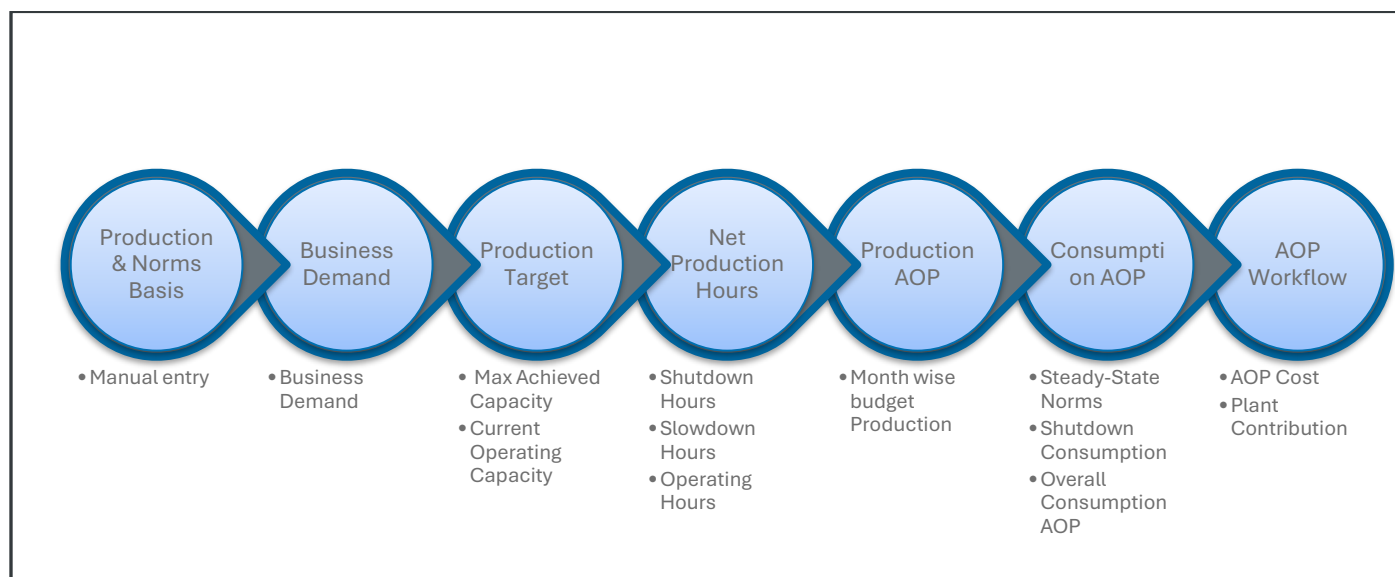


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Production AOP

Below are the steps captured to derive the budgeted AOP production values.

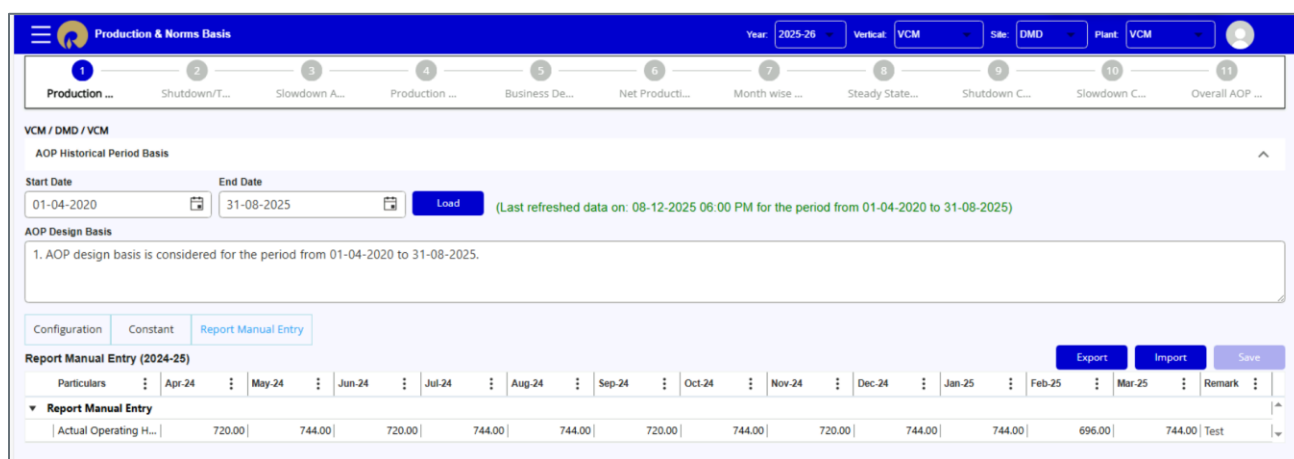
2.1.1.1 Production and Norms Basis

The Production and Norms Basis screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

AOP Design Basis serves as manual entry screen for Highlighting the key insights and summarizing the assumptions using a standard text box feature.

AOP historical period basis added where Start and End date is provided to select historical period to calculate Production Target data and Steady state norms for selected period by default 05 years data will be loaded.

The screenshot below highlights the details captured in the digitized AOP platform.



Production & Norms Basis

Year: 2025-26 Vertical: VCM Site: DMD Plant: VCM

1 Production ... 2 Shutdown/T... 3 Slowdown A... 4 Production ... 5 Business De... 6 Net Producti... 7 Month wise ... 8 Steady State... 9 Shutdown C... 10 Slowdown C... 11 Overall AOP ...

VCM / DMD / VCM

AOP Historical Period Basis

Start Date: 01-04-2020 End Date: 31-08-2025 Load (Last refreshed data on: 08-12-2025 06:00 PM for the period from 01-04-2020 to 31-08-2025)

AOP Design Basis

1. AOP design basis is considered for the period from 01-04-2020 to 31-08-2025.

Configuration Constant Report Manual Entry

Report Manual Entry (2024-25)

Particulars	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25	Remark
Report Manual Entry													
Actual Operating H...	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	696.00	744.00	Test

Figure 2 Digitized AOP screen for Production and Norms Basis

2.1.1.2 Business Demand

The production demand is considered as the maximum value from last year's production values achieved. The default value for VCM is 100%.

The business demand value can be modified by Business group upon providing a specific reason.

AOP digitization portal as seen in the screenshot below.

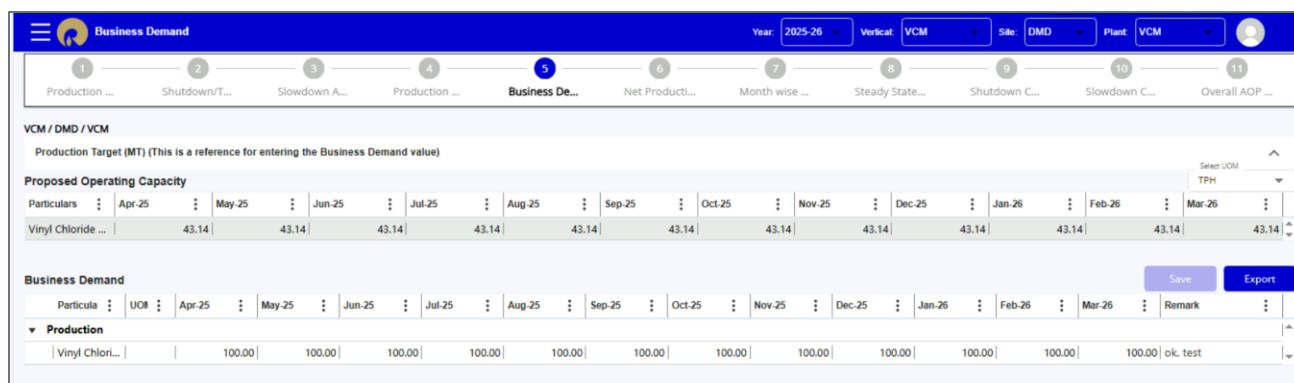


Figure 3 Digitized AOP screen for Business Demand

2.1.1.3 Production Target

Production target screen consists of below grid views.

a) Design Capacity:

Design Capacity is the plant capacity for different furnace modes of operation as per licensors provided data.

b) Max Achieved Capacity:

Max Achieved Capacity is derived as 01 month best from SAP Historical records.

c) Proposed Operating Capacity:

Proposed Operating Capacity will show data similar to Max Achieved Capacity. If the user needs to modify the VCM production rate this can be done by providing a specific reason in remarks column.

This information is captured in the AOP digitization portal as seen in the screenshot below.

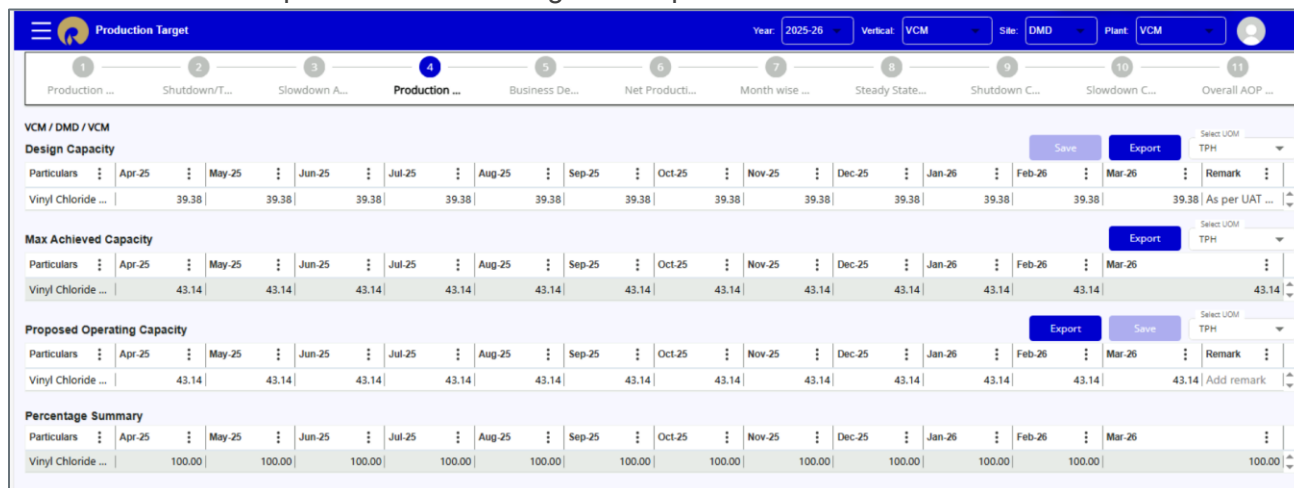
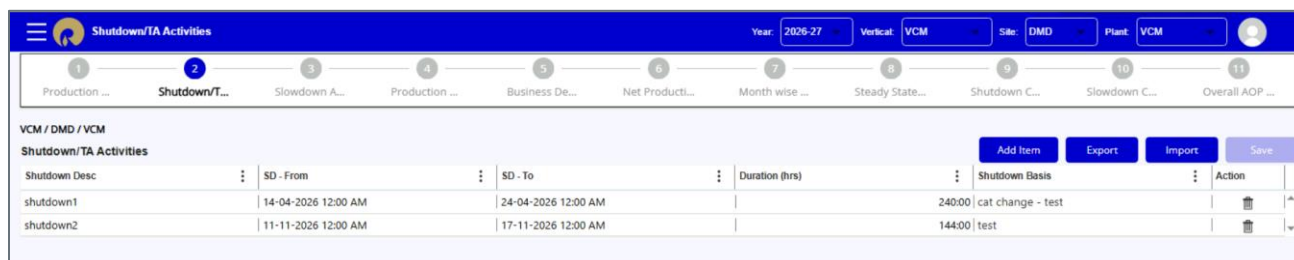


Figure 4 Digitized AOP screen for Production Target

2.1.1.4 Shutdown, Slowdown Activities and Net Production Hours

The operations team will provide the shutdown, turnaround and slowdown plan to calculate month-wise operating hours available for production.

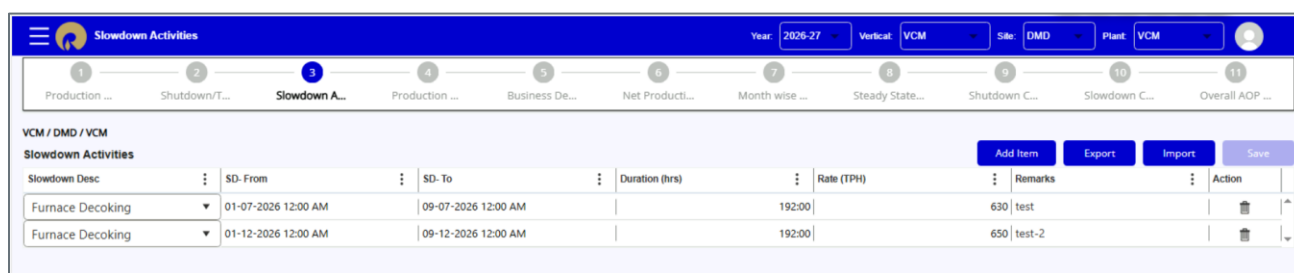
The screenshot below depicts the shutdown and turnaround activities planned, as will be captured in the AOP digitized platform.



The screenshot shows the 'Shutdown/TA Activities' screen in the AOP digitized platform. The interface includes a navigation bar with tabs for Production, Shutdown/TA, Slowdown, Production, Business De..., Net Producti..., Month wise, Steady State, Shutdown C..., Slowdown C..., and Overall AOP. The main content area displays a table of shutdown activities with columns for Shutdown Desc, SD - From, SD - To, Duration (hrs), Shutdown Basis, and Action. Two entries are listed: 'shutdown1' and 'shutdown2'.

Shutdown Desc	SD - From	SD - To	Duration (hrs)	Shutdown Basis	Action
shutdown1	14-04-2026 12:00 AM	24-04-2026 12:00 AM		240.00 cat change - test	[Icon]
shutdown2	11-11-2026 12:00 AM	17-11-2026 12:00 AM		144.00 test	[Icon]

Figure 5 Digitized AOP screen for Shutdown/TA Activities



The screenshot shows the 'Slowdown Activities' screen in the AOP digitized platform. The interface includes a navigation bar with tabs for Production, Shutdown/TA, Slowdown, Production, Business De..., Net Producti..., Month wise, Steady State, Shutdown C..., Slowdown C..., and Overall AOP. The main content area displays a table of slowdown activities with columns for Slowdown Desc, SD - From, SD - To, Duration (hrs), Rate (TPH), Remarks, and Action. Two entries are listed: 'Furnace Decoking' and 'Furnace Decoking'.

Slowdown Desc	SD - From	SD - To	Duration (hrs)	Rate (TPH)	Remarks	Action
Furnace Decoking	01-07-2026 12:00 AM	09-07-2026 12:00 AM		192.00	630 test	[Icon]
Furnace Decoking	01-12-2026 12:00 AM	09-12-2026 12:00 AM		192.00	650 test-2	[Icon]

Figure 6 Digitized AOP screen for Slowdown Activities

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration(hrs) to calculate the SD-To date/time.

The '**Add Item**' button on the Shutdown/Slowdown activities screen allows users to insert a new entry for shutdown/slowdown.

The '**Save**' button enables users to save newly added items as well as any modifications made to existing entries.

The screenshot below depicts the Net Production Hour details captured in the AOP digitized platform.

NOTE- The below screen is read-only. The users cannot change the values.

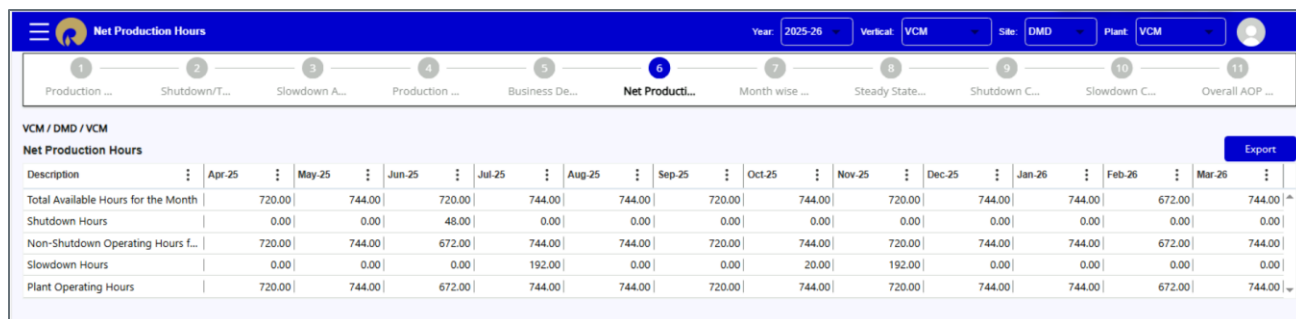


Figure 7 Digitized AOP screen for Net Production Hours

2.1.1.5 Month wise Production plan

Using the Business Demand, Production Target and Maintenance Plan (Operating Hours), the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$VCM (MT) = Proposed Operating Capacity * Operating Hours * (Business Demand/100)$$

$$Operating hours = Total Available hours of the month - Shutdown/TA hours of the month$$

Hence, total budgeted production is,

$$Total Budgeted Production = \sum All the values for the month$$

The month-wise budget production is further used to derive the annual plan in KT, and this is part of the final report.

This information is captured in the AOP digitization portal as seen in the screenshot below.

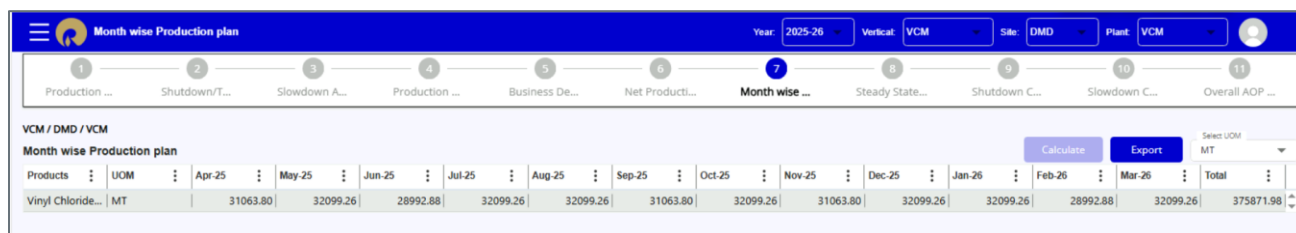


Figure 8 Digitized AOP screen for Month wise Production Plan

The 'Calculate' button on this screen allows users to recalculate Production values when changes are made to production target, business demand, or when line items are added to shutdown activities.

2.1.2 Norms

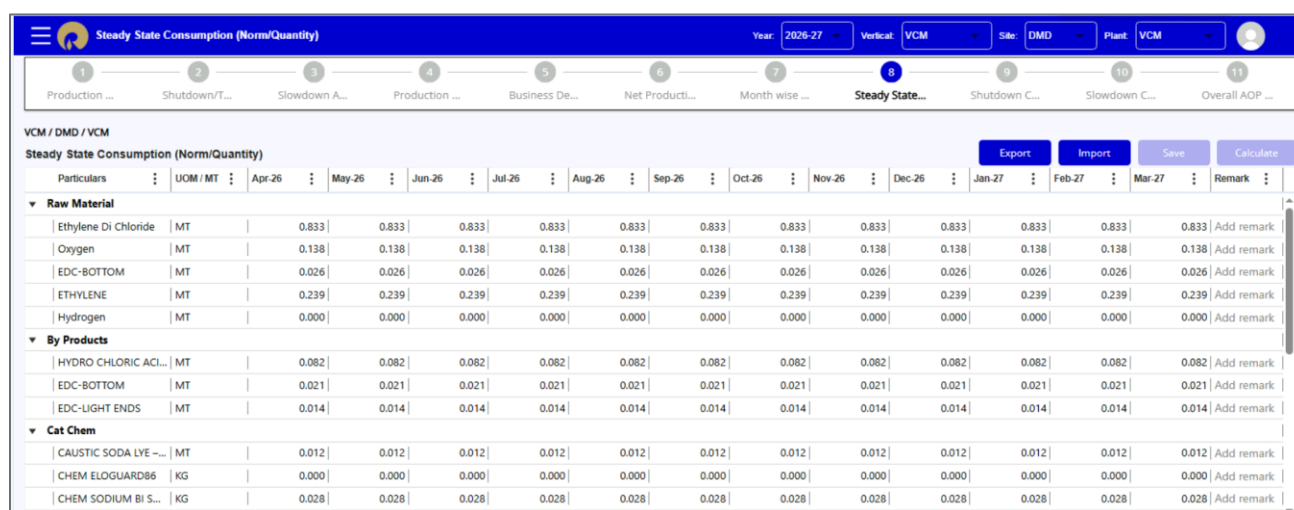
Below are the steps captured to derive the Norms value.

2.1.2.1 Steady State Norms

The Steady state norms screen serves as the basis to calculate steady state norms, allowing users to input essential data for consumption AOP calculations.

Steady state norms are calculated from SAP system by taking Best 03 consecutive months minimum Nos in last 05 years with average Monthly Production is greater than or equal to 211 TPD.

The screenshot below highlights the details captured in the digitized AOP platform in the Steady state norms.



Particulars		UOM / MT	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
Raw Material															
Ethylene Di Chloride	MT		0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	Add remark
Oxygen	MT		0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	Add remark
EDC-BOTTOM	MT		0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	Add remark
ETHYLENE	MT		0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	Add remark
Hydrogen	MT		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Add remark
By Products															
HYDRO CHLORIC ACID	MT		0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	Add remark
EDC-BOTTOM	MT		0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	Add remark
EDC-LIGHT ENDS	MT		0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	Add remark
Cat Chem															
CAUSTIC SODA LYE	MT		0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	Add remark
CHEM ELOGUARD86	KG		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Add remark
CHEM SODIUM BI S...	KG		0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	Add remark

Figure 9 Digitized AOP screen for Steady state Norms

2.1.2.2 Shutdown Norms

Shutdown norms are recorded for the months in which shutdown activities are planned, as specified in the Shutdown/Turnaround Activities screen.

The shutdown quantity utilization data includes TA, asset maintenance linked shutdown/ slowdown activities, etc.

The user must enter the norms data for each line item.

2.1.2.3 Slowdown Norms

Slowdown norms are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The screenshot below highlights the details captured in the digitized AOP platform in the Slowdown norms.

Slowdown Consumption (Norm/Quantity)														
1 2 3 4 5 6 7 8 9 10 11 Production ... Shutdown/T... Slowdown A... Production ... Business De... Net Producti... Month wise ... Steady State... Shutdown C... Slowdown C... Overall AOP ...														
VCM / DMD / VCM Slowdown Consumption (Norm/Quantity) Save Calculate Export														
Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
▼ Raw Material														
Ethylene Di C...	MT	0.000	0.000	0.000	0.891	0.000	0.000	0.000	0.000	0.891	0.000	0.000	0.000	test2
Oxygen	MT	0.000	0.000	0.000	0.141	0.000	0.000	0.000	0.000	0.141	0.000	0.000	0.000	Updated by ...
EDC-BOTTOM	MT	0.000	0.000	0.000	0.038	0.000	0.000	0.000	0.000	0.038	0.000	0.000	0.000	Updated by ...
ETHYLENE	MT	0.000	0.000	0.000	0.250	0.000	0.000	0.000	0.000	0.250	0.000	0.000	0.000	Updated by ...
Hydrogen	MT	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Add remark
▼ By Products														
HYDRO CHLO...	MT	0.000	0.000	0.000	0.121	0.000	0.000	0.000	0.000	0.121	0.000	0.000	0.000	Updated by ...
EDC-BOTTOM	MT	0.000	0.000	0.000	0.038	0.000	0.000	0.000	0.000	0.038	0.000	0.000	0.000	Updated by ...
EDC-LIGHT E...	MT	0.000	0.000	0.000	0.018	0.000	0.000	0.000	0.000	0.018	0.000	0.000	0.000	Updated by ...
▼ Cat Chem														
CAUSTIC SOD...	MT	0.000	0.000	0.000	0.018	0.000	0.000	0.000	0.000	0.018	0.000	0.000	0.000	Updated by ...
CHEM ELOGU...	KG	0.000	0.000	0.000	0.004	0.000	0.000	0.000	0.000	0.004	0.000	0.000	0.000	Updated by ...
CHEM SODIU...	KG	0.000	0.000	0.000	0.061	0.000	0.000	0.000	0.000	0.061	0.000	0.000	0.000	Updated by ...

Figure 10 Digitized AOP screen for Steady state Norms

2.1.2.4 Budget Norms/Consumption AOP

Using the values of Steady state norms and Shutdown Consumptions the final Consumption AOP value is calculated as below.

$$\text{Consumption AOP for months without shutdown} = \text{Steady State Norms}$$

$$\text{Consumption AOP for months with shutdown} = \text{Steady State Norms} + \text{Shutdown Norms}$$

3. AOP Reports

Following report will be implemented as part of digital AOP workflow.

- Annual AOP Cost
- Plant Production Summary
- Month wise Production Plan
- Month wise raw data
- Turnaround report
- Annual Production Plan
- Plant Contribution (T-21)
- Plant Contribution Summary (T-22)

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by VMD Site team